Research Brief: Ontology-Guided Synthetic Data Generation for a Biomedical Track A Project (ENGG 119, Fall 2026)

TL;DR

- **The project's strongest, most defensible spine is a narrow one: use a biomedical ontology to detect under-represented regions of a small clinical tabular dataset, fill those gaps with an LLM-based generator (prompt-based CLLM or fine-tuned GReaT), and prove it beats CTGAN/TVAE and SMOTE on a downstream rare-class detection task using standard TSTR + fidelity metrics.** This is achievable in 10 weeks, has clear published precedent (CLLM, ICML 2024), and produces a clean quantitative result.
- **Do NOT attempt both the tabular GAN/VAE comparison AND SLM-beats-frontier-LLM distillation as co-equal thrusts.** The SLM-distillation claim is real in the literature (e.g., Meerkat-7B passed USMLE; Mistral-7B QLoRA beat GPT-4o/GPT-5 on biomedical claim verification at 44.5× lower cost) but reproducing it on a *narrow* benchmark in 10 weeks with three students is high-risk and should be a clearly-labeled stretch goal or the Track B follow-on. Pick the tabular thrust as primary.
- **"Bayesian knowledge-driven ontologies" is not a standard term; the real literature is PR-OWL / Multi-Entity Bayesian Networks (MEBN), probabilistic ontologies, and — more practically for a student team — ontology/SHACL-based coverage analysis plus knowledge-graph completion.** The team should reframe "Bayesian knowledge-driven ontology" as "ontology-guided coverage analysis with a probabilistic gap score" to avoid overclaiming a heavyweight formalism they cannot implement.

A. Ontology-Guided Gap Detection and Bayesian Methods

A1. What "Bayesian knowledge-driven ontologies" actually means in the literature

The phrase as written is not a recognized term of art. The closest established bodies of work are:

- **PR-OWL (Probabilistic OWL)** and its successor **PR-OWL 2**, which extend the OWL web ontology language with formal probabilistic semantics. PR-OWL is built on **Multi-Entity Bayesian Networks (MEBN)**, a first-order probabilistic logic that unifies Bayesian probability with first-order logic (Costa & Laskey, FOIS 2006). In MEBN, probabilistic knowledge is expressed as **MEBN fragments (MFrags)** organized into **MEBN Theories (MTheories)** that guarantee a unique joint probability distribution over random variables. Upon query, MEBNs produce **Situation-Specific Bayesian Networks (SSBNs)**.
- **BayesOWL**, an earlier approach that extends OWL using Bayesian networks as the underlying reasoning layer.
- A known, candidly-stated limitation: PR-OWL 2 is built on OWL 2 DL (description logic SROIQ(D)) and has **high computational complexity / scalability problems**, and PR-OWL/MEBN have historically "lacked a software tool that implements their underlying concepts" (multiple survey papers 2008–2017). This is a red flag for a 10-week implementation.

Concretely, "putting a probability distribution over an ontology's classes and relations" means: for each class/relation, representing uncertainty about whether an instance belongs to a class or whether a relation holds, via conditional probability tables attached to ontology fragments (MFrags). This is heavyweight. **Recommendation: the team should not implement MEBN/PR-OWL.** Instead, reframe "Bayesian" as a lightweight probabilistic *coverage/gap score* over ontology concepts (e.g., expected vs. observed instance counts per concept, with a Bayesian/Dirichlet smoothing prior). Cite PR-OWL/MEBN as the formal backdrop they are *inspired by* but deliberately simplifying.

A2. Formally detecting "gaps" / incompleteness relative to an ontology

Several concrete, implementable threads exist:

- **SHACL (Shapes Constraint Language)** and **ShEx** for ontology-based data validation. SHACL operates under a **closed-world assumption** ("what is not stated is false"), which is exactly what you want for completeness checking, whereas OWL uses the **open-world assumption**. SHACL core can be used to compute data-quality metrics: per "Is SHACL Suitable for Data Quality Assessment?" (arXiv 2507.22305), which defined SHACL shapes for the 69 DQ metrics of Zaveri et al. (2012): "Seven dimensions present a coverage between 50-90%: Timeliness, Understandability, Consistency, Completeness, Syntactic Validity, Interoperability, Interpretability, and Versatility," while "SHACL can check the usage of classes, it cannot check property usage." Caveat from the literature: SHACL core can check *class* usage but not always *property* usage, and "cannot detect wrong values in a plausible range" (ABECTO paper) — i.e., validation finds missing/invalid fields, not subtle distributional gaps.
- **Knowledge-graph completion / link prediction** for inferring missing edges.

- **Coverage/enrichment analysis**: comparing observed instance counts per ontology concept against expected distributions to flag under-represented concept-space regions. This is the most student-friendly operationalization of "gap detection."
- **Open-world vs. closed-world assumption** is the key conceptual pivot: completeness checking requires temporarily adopting CWA (what SHACL does).

Recommended operational definition for the proposal: a "gap" = an ontology concept (e.g., an HPO phenotype or a MONDO disease subtype) that is present in the ontology but has zero or statistically under-represented support in the dataset relative to a prior (ontology-weighted expectation). This is implementable with OLS/BioPortal APIs + pandas + a simple SHACL/count check.

A3. Prior work on ontology/KG-guided synthetic data generation in biomedicine

This is a live, credible research area — good for the team because it shows novelty without being unprecedented:

- **BioGAN** (Panaccione et al., *Bioengineering* 2025, DOI 10.3390/bioengineering12060658): first to incorporate graph neural networks into a GAN for transcriptomic data, using gene-regulatory/co-expression networks to preserve biological plausibility; validated on *E. coli* and human gene expression. Finding: incorporating a priori biological knowledge improves both quality and utility of synthetic data.
- **"Novel Knowledge-Guided Generative Methods for Synthetic Transcriptomic Data"** (arXiv 2608.13256) — proposes Graph-Modulated GAN (GM-GAN) and Graph-Regularized GAN (GR-GAN) that inject biological knowledge graphs, and finds graph-informed models outperform unconditional ones.
- **KIPPS** (arXiv 2409.17315) — "Knowledge Infusion in Privacy Preserving Synthetic Data Generation," uses knowledge graphs to impose domain constraints so generated data obeys rules (demonstrated in cybersecurity/healthcare framing).
- **OntoBook** (arXiv 2607.18927) — converts medical ontology structure into pretraining signal by generating random walks through ontology graphs and reformulating them into synthetic "textbook" prose via an LLM. This is a direct, citable precedent for the team's "generate data that fits the ontology" idea.
- Biomedical KG resources that ground such work: **Hetionet, PrimeKG, RTX-KG2, the Monarch KG, the Biolink Model**.

Takeaway: the ontology-guided generation idea is defensible and has 2024–2026 precedent, but existing work is mostly in transcriptomics/omics (high-dimensional) rather than simple clinical tabular data. The team's contribution can honestly be "apply ontology-guided gap-filling to a clinical tabular detection task and benchmark it against standard generators."

B. Synthetic Data Generation — LLM vs GAN vs VAE

B4. Baselines and the LLM-vs-GAN/VAE performance picture

The baselines to compare against (all in the Synthetic Data Vault / synthcity ecosystems):

- **GAN-based:** CTGAN, CopulaGAN, CTAB-GAN / CTAB-GAN+, medGAN, ADS-GAN.
- **VAE-based:** TVAE.
- **Statistical:** Gaussian Copula, synthpop, Bayesian networks (CLBN/PrivBN).
- **Diffusion:** TabDDPM, TabSyn.
- **LLM-based: GReaT** ("Language Models are Realistic Tabular Data Generators," Borisov et al., arXiv 2210.06280), Distill-GReaT, TabuLa (Zhao et al. 2025), TapTap, HARMONIC (NeurIPS 2024), and **CLLM** (Curated LLM, Seedat et al., ICML 2024).

What the literature says about relative performance:

- GReaT reported it "decreases the discriminator performance by 16.2% over other competitive baselines" (i.e., its synthetic rows are harder to distinguish from real). But GReaT is **expensive**: the original paper reports training/fine-tuning time of ~9h10m for GReaT vs. 0:46 for TVAE and 1:10 for CTGAN, and sampling 17s vs. 0.045s for CTGAN. This cost matters for a student timeline.
- A GPT-4o zero-shot study (Researchgate 389176619, "GANs vs LLMs… synthetic tabular") found **GPT-4o outperformed CTGAN in preserving means, 95% CIs, bivariate correlations, and privacy** even without fine-tuning — but with the important failure mode that LLM output violated "definitional ranges (e.g., negative measurements)."
- **Crucially for this project, the low-data regime is where LLM-based methods shine.** CLLM (ICML 2024) "empirically demonstrate[s] on 7 real-world datasets that CLLM enables superior downstream performance compared to 6 widely used tabular data generative models" (including CTGAN, TVAE, TabDDPM, SMOTE, GReaT), with gains "especially noticeable in the low-data regime (n<100)" and "for underrepresented subgroups." This is the single most important finding for the team — it directly supports the thesis and is reproducible (code at github.com/seedatnabeel/CLLM).
- Contrary caution: "Why LLMs Are Bad at Synthetic Table Generation" (arXiv 2406.14541) argues in-context LLM generation has real failure modes and proposes fine-tuning fixes — the team should cite this for balance.
- A prosumer-hardware benchmark (CODATA Data Science Journal 2025) gives head-to-head CTGAN/TVAE/GReaT numbers on imbalance, augmentation, privacy (DCR), and ML utility — a good methodological template.

B5. Standard, implementable evaluation metrics (three buckets)

This is well-standardized and very doable in 10 weeks. Use these libraries: **SDMetrics** (DataCebo), **synthcity** (van der Schaar lab), **SynthEval**, **TableEvaluator**, **Anonymeter** (privacy).

- **(a) Fidelity:** Kolmogorov–Smirnov statistic (numerical marginals), Total Variation Distance (categorical), pairwise correlation difference, Hellinger distance, Jensen–Shannon / KL divergence, Wasserstein distance, **propensity-score MSE (pMSE)** (range [0, 0.25)), and **discriminator/detection AUC** (how well a classifier separates real vs. synthetic — lower = better).
- **(b) Utility: Train-on-Synthetic-Test-on-Real (TSTR)** vs **Train-on-Real-Test-on-Real (TRTR)**; downstream classifier accuracy/AUC/F1. For imbalanced tasks, prefer **AUC-PR / macro-F1** over accuracy. A representative colorectal-cancer study (PMC11330441) reports TSTR/TRTS AUC of 0.99/0.98 and pMSE 0.223 — a good benchmark for "what good looks like."
- **(c) Privacy: Distance to Closest Record (DCR)**, Nearest-Neighbor Distance Ratio (NNDR), **membership-inference** risk, and (if ambitious) **differential-privacy** guarantees (CTAB-GAN+ and DP-SGD support this).

Recommendation: SDMetrics + synthcity cover all three buckets out of the box and are pip-installable; this is the safest evaluation stack. Budget time for the known annoyance that SDMetrics aggregates metrics into single scores (limiting per-metric comparison) — SynthEval is more granular.

B6. Known failure modes and mitigations

- **Model collapse / recursive training on synthetic data:** Shumailov et al., *Nature* 631:755–759 (2024), "AI models collapse when trained on recursively generated data" (DOI 10.1038/s41586-024-07566-y) — training successively on synthetic output causes loss of distribution tails and eventual degeneration. **Mitigation (well-supported):** always mix real + synthetic rather than training purely on synthetic; Gerstgrasser et al. (2024) show **accumulating** real data with synthetic avoids collapse. This directly justifies the team's "keep a human-expert / real-data loop" design.
- **Mode collapse in GANs:** CTGAN's conditional vector + "packing" and CTAB-GAN+'s log-probability conditional vector specifically target this for imbalanced data.
- **LLM hallucination producing clinically implausible records:** e.g., negative lab values, impossible code combinations. **Mitigation:** ontology/SHACL post-generation validation (this is a natural selling point for the ontology-guided approach) and CLLM-style **curation** (filter generated rows by predictive confidence / aleatoric uncertainty).
- **Memorization / leakage:** measure with DCR and membership inference; report it as a feature, not an afterthought.

C. SLM Distillation and Benchmarks

C7. Can a small model fine-tuned on domain (synthetic) data beat a frontier model on a narrow benchmark?

The evidence that this *can* happen is strong — but the student-feasibility is the catch.

- **Meerkat-7B/8B** (*npj Digital Medicine* 2025, s41746-025-01653-8): fine-tuned on chain-of-thought extracted from 18 medical textbooks (441K examples). **Meerkat-7B scored 70.6% (single) / 77.1% (ensemble) on MedQA — the first 7B model to pass the USMLE 60% threshold** — beating MediTron-70B's 70.2% and GPT-3.5. Meerkat-70B (82.6%) surpassed GPT-4 by 1.2%. Fine-tuning used ~28× less GPU time than continued pretraining.
- **Mistral-7B QLoRA on biomedical claim verification** (arXiv 2606.12854): achieved **88.4% macro-F1 on SciFact and 65.2% on HealthVer, beating GPT-4o (85.6%, 53.2%) and GPT-5 (77.9%, 42.4%) at 44.5× lower cost using just 1,008 training examples.** This is the single best "SLM beats frontier at low cost with few examples" citation for the proposal.
- **Phi / "Textbooks Are All You Need"** (Gunasekar et al., arXiv 2306.11644): phi-1 (1.3B) hit 50.6% pass@1 on HumanEval trained on 6B tokens of textbook-quality + 1B tokens of GPT-3.5-generated synthetic data — the foundational "synthetic data lets small models punch above their weight" result.
- **MEDITRON-70B** (arXiv 2311.16079) and **BioMistral-7B** as domain baselines.

Contrary / sobering evidence: On broad clinical tasks, open models still trail frontier ones — "Towards evaluating… versatile LLMs for medicine" (*npj Digital Medicine* 2024) shows MEDITRON at 68.27 vs GPT-4's 84.73 and Claude-3.5's 92.93 on SEER treatment planning. And a construct-validity study (arXiv 2503.10694) shows MedQA scores do **not** correlate well with real-world EHR performance — so "beating a benchmark" ≠ clinical utility. **Honest assessment: the "SLM beats frontier LLM" result is real but was achieved by well-resourced labs; a three-student team reproducing it on a *custom* narrow benchmark in ~6 usable weeks is high-risk.** Make it the stretch goal.

C8. Benchmarks — accessibility, licensing, harnesses

- **MedQA (USMLE):** 1,273-question test set, 4-option English variant widely used; training set GBaker/MedQA-USMLE-4-options (10,178 examples) is on Hugging Face. Easy to run. Passing threshold 60%.
- **MedMCQA:** ~194K questions (AIIMS/NEET); evaluate on the **4,183-question validation set** since test answer keys are not public.
- **PubMedQA:** 500 expert-labeled yes/no/maybe items with abstract context. Easy.
- **MMLU clinical subsets** (anatomy, clinical knowledge, college biology/medicine, medical genetics, professional medicine): no training set, good for cross-domain transfer testing.
- **BioASQ** (factoid QA), **HealthBench**, **MedXpertQA** — more specialized/heavier.

- **Harnesses: EleutherAI lm-evaluation-harness** (the de facto standard) and **HELM**.
- **Reference scores (as of 2026):** frontier models now score ~95% on MedQA (o4-mini-high 95.2%, Gemini 2.5 Pro 94.6%, DeepSeek R1 92.1% per the LayerLens/pricepertoken leaderboard, Oct 1 2026); best sub-10B open models with fine-tuning reach ~70–77% (Meerkat-7B). **The realistic student win is not "beat the frontier model on MedQA" (nearly impossible now) but "on a narrow, self-defined slice, a fine-tuned small model closes most of the gap to a frontier model at a fraction of cost."**

C9. Practical fine-tuning tooling and compute cost

Very favorable for a student team:

- **Methods:** LoRA / **QLoRA** (4-bit NF4), with **Unsloth** (fastest, lowest-VRAM), **Axolotl** (most configurable), and Hugging Face **TRL/PEFT**.
- **Base models (0.5B–8B):** Qwen (2.5/3), Llama 3.1-8B, Gemma, Phi, SmolLM. Open models now ship 128K context and are "competitive with generic big ones on narrow tasks."
- **Compute cost (2026 figures):** a 7–8B QLoRA fine-tune needs only **6–10 GB VRAM**, runs in **2–4 h on an A100** or **6–8 h on an RTX 4090**, and costs **~**$3–$**10 on a rented A100 or under ~**$3 on a RunPod RTX 4090 (~**0.34/hr)**. On a free Colab T4, small fine-tunes (1k–10k examples, 7B) finish in **15–60 minutes** with Unsloth. This is well within a student budget.

Takeaway: compute is not the bottleneck; data quality, evaluation rigor, and scope are.

D. Disease-Detection / Traditional ML Angle

D10. Accessible clinical/biomedical tabular datasets (and the schedule risk)

Immediately usable, no credentialing (use these for the core demo):

- **UCI Heart Disease (Cleveland)** — classic 303-record binary/multiclass detection task, naturally small (good for low-data story).
- **Pima Indians Diabetes** — classic small imbalanced binary task.
- **UCI Diabetes 130-US Hospitals** — larger, used in recent synthetic-data audit benchmarks.
- **NHANES** — openly available national survey data.
- **Synthea / SyntheticMass** — fully open, no restrictions; ~1M synthetic patients, SNOMED-CT/RxNorm/LOINC-coded, models the 10 most common primary-care reasons and 10 highest-morbidity chronic conditions. **Ideal for prototyping before real data clears.** Honest caveat: Synthea has documented fidelity flaws (e.g., it "incorrectly

generated 20% of patients in the infant age group" and "failed to note diabetes onset after age 52" per the JAMIA 2018 paper), so it's a dev/prototyping substrate, not a gold-standard evaluation set.
- **MIMIC-IV demo** and **eICU demo** — the demo subsets are open and require **no credentialing or CITI training**.

Credentialed (SCHEDULE RISK — start week 1 if needed):

- Full **MIMIC-IV / MIMIC-IV-Note / MIMIC-IV-ED** on PhysioNet require: PhysioNet account → credentialed-user application (requires a reference) → **CITI "Data or Specimens Only Research"** training → sign the dataset-specific DUA. **Approval time is normally a few days to 1–2 weeks, but PhysioNet has warned of delays "up to 45 days" during high-volume periods.** Each dataset needs its own DUA.

Recommendation: build the entire pipeline on UCI/Pima/Synthea (open) so the project never blocks on credentialing; if a team member wants real EHR data, start the MIMIC-IV credentialing on day one as a parallel, non-blocking track.

D11. Does synthetic augmentation actually beat SMOTE/ADASYN/class-weighting? (Be honest: mixed.)

The evidence is genuinely mixed, and the proposal should say so:

- **Pro-GAN findings:** an omics-data study (PMC10979623) found GAN-based augmentation beat SMOTE and random oversampling on small high-dimensional data; a cancer-prediction study (PMC11640389) raised average ROC-AUC from 0.8276 to >0.9734 with GAN resampling.
- **Pro-simpler-baseline findings:** the survey "The use of generative adversarial networks to alleviate class imbalance in tabular data" (*Journal of Big Data* 2022, 10.1186/s40537-022-00648-6) notes "83% of unique experiments among our researched papers boasted evaluation metrics for machine learning efficacy results where experimental methods outperformed baseline methods" — i.e., GANs beat SMOTE/ADASYN in most but not all cases, confirming the mixed picture; it also reports cases where "GAN-based methods yielded classifiers which were no more performant… than SMOTE and ADASYN," and sometimes **no resampling at all was optimal**. A GAN-vs-SMOTE/ADASYN study (arXiv 1904.09135) found the GAN "improved results… but did not perform better than SMOTE or ADASYN."
- **The nuanced consensus:** no method dominates; for high-dimensional, limited-sample, privacy-sensitive settings, classical copula/synthpop often give the best utility-privacy tradeoff (PMC12492482, pharmacogenetics). And LLM-based CLLM specifically claims superiority **in the low-data regime** where SMOTE struggles (SMOTE interpolates between existing minority points and cannot extrapolate beyond them).

This mixed evidence is an asset, not a liability: it makes the team's head-to-head comparison (LLM/ontology-guided vs. GAN/VAE vs. SMOTE vs. class-weighting vs.

no-augmentation) a genuine, publishable question rather than a foregone conclusion. **The proposal must include SMOTE, ADASYN, class-weighting, and a no-augmentation baseline** — omitting them is the most common reviewer criticism.

E. Business / Track A Framing

E12. The commercial landscape (who could be "reimagined with AI at its core")

Synthetic health data is a real, funded market with recent M&A:

- **Gretel.ai** — founded 2019; raised ~$67M (Greylock, Anthos; Series B $52.2M, Oct 2021 ata$ ~320M valuation); **acquired by NVIDIA, reported by Wired on March 19, 2025 (citing "two people with knowledge of the deal"), for a price that "came in at nine figures and was higher than the $320 million at which Gretel was most recently valued," with "roughly 80 employees" folded into NVIDIA and neither company issuing a release.** Product: synthetic tabular/text/time-series with tunable differential privacy.
- **MDClone** (Israel, founded 2016) — self-service health-data "ADAMS" platform with a patented Synthetic Data Engine; **raised $104M total; $63M Series C (Mar/Apr 2022)** led by Warburg Pincus & Viola Growth. Clients include Washington University, Intermountain, Regenstrief. **Best "existing product to reimagine" candidate** because it's a health-specific, workflow-centric product.
- **MOSTLY AI** (Vienna, founded 2017) — raised ~$31M; **$25M Series B (Jan 2022)** led by Molten Ventures ("Europe's largest synthetic-data investment"); released an open-source Synthetic Data SDK (Feb 2025); the **brand was acquired by Syntho, confirmed by a Syntho press release (June 9, 2026): "Syntho … today announced the acquisition of the MOSTLY AI Brand. Going forward, the MOSTLY AI brand will continue under the name 'MOSTLY AI, powered by Syntho,'" with Syntho acquiring the trademark and related assets** after the company wound down operations.
- **Tonic.ai** (SF, founded 2018) — raised **$45M; $35M Series B (Sep 2021)** led by Insight Partners; developer-focused synthetic/de-identified test data; publishes HIPAA Expert Determination guidance as marketing.
- **Replica Analytics** (Ottawa; CEO Dr. Khaled El Emam) — **acquired by Aetion, announced Jan 4, 2022, terms undisclosed**; "Replica Synthesis" generates privacy-protected synthetic copies of real-world data; now part of Aetion's regulatory-grade RWE platform.
- **Hazy** (London, founded 2017) — raised ~$9–15M; **principal software assets acquired by SAS (Nov 2024, terms undisclosed)**, folded into "SAS Data Maker."
- **Syntegra** (SF, founded 2019) — $5.625M seed (Jan 2022); "Medical Mind" generated synthetic versions of NIH N3C COVID data; **current status ambiguous (reported out-of-business 2023 or M&A 2025)** — verify before citing.

- **Betterdata** (Singapore, founded 2021) — ~$1.55M seed (Apr 2023, Investible-led); programmable synthetic-data SaaS.
- **Synthea** — open-source (MITRE); the free baseline the team should actually use.
- **SYNTHEMA** — note: this is **not a company** but an **EU Horizon Europe research project (grant 101095530, ~€7M, 16 partners)** on AI synthetic data for rare haematological diseases. ("Simulacra" as a vendor could not be verified — likely a naming confusion.)

E13. Market size and regulatory drivers

Market size (note: estimates vary widely by analyst — report the range honestly):

- Grand View Research (grandviewresearch.com): the global synthetic data generation market was "valued at USD 218.3 million in 2023 and is projected to grow from USD 528.3 million in 2026 to USD 1,788.1 million by 2030, at a CAGR of 35.3% from 2024 to 2030"; North America held 34.5% (2023) and tabular data 38.8% of revenue, and healthcare & life sciences is the largest/dominant end-use segment.
- MarketsandMarkets (via MathCo): **$0.3B (2023) → $2.1B (2028), 45.7% CAGR**.
- Grand View, synthetic-data-in-healthcare specifically: various analysts put 2025 at **~$500–660M** growing at **~24–31.5% CAGR** through the early-mid 2030s (DataM Intelligence: $657.92M in 2025 → $5.88B by 2033). **These are consultancy estimates; treat as order-of-magnitude, not precise.**

Regulatory drivers (the actual business pain):

- **HIPAA de-identification** has two routes: **Safe Harbor** (remove 18 enumerated identifiers; mechanical, destroys utility, doesn't handle quasi-identifiers well for rare conditions) and **Expert Determination** (45 CFR §164.514(b)(1); a qualified expert certifies "very small" re-identification risk; preserves utility, allows date-shifting). Synthetic data is marketed as a way to sidestep both. For **rare diseases/small populations, Safe Harbor is inadequate** because residual quasi-identifiers re-identify people — a direct selling point for synthetic data.
- **GDPR** (EU) applies to all personal data of EU data subjects; synthetic data that is truly non-personal falls outside it.
- **EU AI Act** data-governance provisions (Article 10) require high-quality, representative training data for high-risk systems — synthetic data is cited as a mitigation.
- **FDA:** In January 2025 FDA issued its **first draft guidance on AI in drug/biologic development — "Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products"** (Docket FDA-2024-D-4689), built around a **7-step, risk-based credibility-assessment framework** tied to a model's "context of use." **Synthetic data is NOT substantively addressed** in the draft — an FDA/CTTI workshop report explicitly flags "guidance on the use of synthetic data" as a future policy gap. FDA separately issued **"Artificial Intelligence-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations"** (Docket FDA-2024-D-4488, Jan 2025), a

total-product-lifecycle framework (FDA has authorized >1,000 AI-enabled devices). FDA's **Real-World Evidence** program stems from the **21st Century Cures Act (2016)**; the **"Framework for FDA's Real-World Evidence Program" (Dec 2018)** defines RWD/RWE but does not treat synthetic data as an RWD source. **Why firms buy synthetic data:** IRB/DUA delays, data-access cost, privacy liability, and rare-disease data scarcity.

E14. Who is the buyer / user?

Candidate personas and procurement paths:

- **Pharma clinical data scientists** — highest willingness to pay; buy via enterprise procurement/MSA; pain = trial data scarcity, RWE generation, cross-border data sharing.
- **Health-system ML engineers** — moderate budget; pain = IRB/governance delays accessing their own EHR; MDClone's actual customer.
- **Digital-health startups without data access** — high need, low budget; pain = cold-start with no data; want cheap/open tools.
- **Medical-device makers** — buy validation data to support FDA submissions; pain = need representative edge-case/rare-condition data.
- **EHR vendors & academic researchers** — researchers are low-WTP but high-volume; academic/open procurement.

Recommended Track A persona for the proposal: a **digital-health-startup or academic ML engineer facing a rare/under-represented clinical class with too little data and no fast path to more** — this persona makes the ontology-guided gap-filling value proposition concrete and matches a product (not research) framing.

F. Scope and Feasibility Critique

F15. Which of the five bullets are achievable in 10 weeks by three students?

Frank assessment of each verbatim bullet:

1. **Ontology-guided gap detection + LLM fills gaps to fit ontology — ACHIEVABLE and should be the core.** Use OLS/BioPortal + a coverage/gap score + CLLM/GReaT generation + SHACL/ontology validation.
2. **Ontology identifies gaps → go back to human expert for real data; generate questions for an LLM to fill gaps — ACHIEVABLE as a UI feature** (human-in-the-loop); this doubles as the "clinically plausible / avoid model collapse" safeguard. Keep it lightweight.

3. **Generate data to train a traditional ML model (disease detection) — ACHIEVABLE and should be the evaluation backbone** (TSTR on UCI/Pima/Synthea).
4. **Use synthetic data to distill an SLM that beats the original SOTA LLM on a narrow benchmark — HIGH RISK; defer to stretch goal or Track B.** The literature says it's possible but the engineering + evaluation load is too much on top of bullets 1–3.
5. **Compare LLM synthesis vs autoencoders (VAE) vs GANs — ACHIEVABLE and should be the headline comparison** (CLLM/GReaT vs TVAE vs CTGAN/CTAB-GAN vs SMOTE/ADASYN/class-weighting/no-aug), scored with SDMetrics/synthcity.

Is it realistic to do BOTH the tabular GAN/VAE comparison AND the SLM distillation? No. Each is a full project. **Recommended primary thrust:** ontology-guided gap detection → LLM gap-filling → head-to-head vs GAN/VAE/SMOTE → downstream rare-class detection (bullets 1, 2, 3, 5). **Recommended stretch goal:** if ahead of schedule, distill a small model on the best synthetic data and show it narrows the gap to a frontier LLM on one narrow self-defined task (bullet 4). **Track B follow-on:** frame the SLM-distillation + an agentic "data-gap-filling copilot" as the internal-workflow extension.

F16. A concrete, minimal, visual Demo-Day architecture

Track A requires a *product*, so the demo must be clickable, not a notebook. Proposed end-to-end flow (buildable with Streamlit/Gradio front-end + Python back-end + an LLM API):

1. **Upload / pick a dataset** (default: UCI Heart Disease or a Synthea extract). User sees row count and class balance.
2. **"Analyze gaps"** — the app maps columns to an ontology (HPO/MONDO/SNOMED via OLS/BioPortal API) and renders a **visual coverage map / tree** highlighting under-represented concepts in red (the "Bayesian coverage score"). *This is the visual wow-moment and the ontology differentiator.*
3. **"Fill gaps"** — user picks a method from a dropdown (**CLLM / GReaT / CTGAN / TVAE / SMOTE**) and the app generates synthetic rows targeting the red regions; ontology/SHACL validation rejects implausible rows live (show the rejected-row counter).
4. **"Human-in-the-loop"** (optional tab) — for the thinnest gaps, the app auto-drafts questions for a human expert/LLM to supply real values.
5. **"Evaluate"** — one-click dashboard: fidelity (KS/TVD, correlation heatmaps), **TSTR vs TRTR AUC/F1 bar chart**, and privacy (DCR). The headline number: **rare-class recall/AUC improvement vs. the no-augmentation baseline.**
6. **Download** the augmented dataset + a one-page "synthetic data quality report."

What the judge sees and clicks: upload → a red/green ontology coverage tree lights up → pick a generation method → watch gaps fill and implausible rows get rejected → a dashboard shows the downstream detection model getting measurably better. That is a product narrative, not a research artifact.

Key Papers Worth Citing in the Proposal

1. **Seedat et al., "Curated LLM (CLLM)…," ICML 2024** (arXiv 2312.12112) — LLM synthetic data beats CTGAN/TVAE/TabDDPM/SMOTE/GReaT in low-data regimes and for under-represented subgroups. *The anchor citation.*
2. **Borisov et al., "Language Models are Realistic Tabular Data Generators (GReaT)," ICLR 2023** (arXiv 2210.06280).
3. **Xu et al., "Modeling Tabular Data using Conditional GAN (CTGAN/TVAE)," NeurIPS 2019** (arXiv 1907.00503).
4. **Shumailov et al., "AI models collapse when trained on recursively generated data," Nature 2024** (10.1038/s41586-024-07566-y) — justifies the real-data-in-the-loop design.
5. **Kim et al., "Small language models learn enhanced reasoning skills from medical textbooks (Meerkat)," npj Digital Medicine 2025** (s41746-025-01653-8).
6. **Gunasekar et al., "Textbooks Are All You Need (phi-1)," 2023** (arXiv 2306.11644).
7. **Costa & Laskey, "PR-OWL: A Framework for Probabilistic Ontologies," FOIS 2006** — the formal "Bayesian ontology" backdrop.
8. **Panaccione et al., "BioGAN," Bioengineering 2025** (10.3390/bioengineering12060658) — ontology/KG-guided biomedical generation.
9. **CODATA Data Science Journal 2025, "Benchmarking Tabular Data Synthesis… on Prosumer Hardware"** — methodological template for the comparison.
10. **Walonoski et al., "Synthea," JAMIA 2018** (25(3):230) — the open dataset the team should actually use.

Caveats and Weak/Contested Evidence (flagged honestly)

- **"Bayesian knowledge-driven ontology"** is not standard terminology; PR-OWL/MEBN is the real thing but is too heavyweight to implement — reframe as lightweight probabilistic coverage scoring.
- **Does synthetic augmentation beat SMOTE?** Genuinely mixed; sometimes no augmentation wins. Must include simple baselines.
- **SLM-beats-frontier claims** are real in the literature but achieved by resourced labs; MedQA scores also have weak construct validity vs. real EHR outcomes — do not overclaim clinical significance.
- **Market-size numbers** are consultancy estimates that vary 2–3× between analysts; cite as ranges.
- **Several vendor statuses/acquisition prices are undisclosed or ambiguous** (Gretel's price "nine-figure"/undisclosed; Syntegra's status; MOSTLY AI's wind-down) — cite cautiously.
- **FDA has not issued synthetic-data-specific guidance**; it is explicitly flagged as a policy gap — don't imply an approval pathway exists.

- **Frontier models now score ~95% on MedQA**, so "beat the frontier LLM on a public medical benchmark" is effectively off the table for a student team; scope the SLM claim to a narrow, self-defined slice and a cost/efficiency framing.